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RESEARCH PAPER

Designing multilayer mirror structures to achieve high pupil transmittance uniformity in EUV lithography

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ABSTRACT. The purpose of this work is to develop design schemes for extreme ultraviolet (EUV) multilayer mirrors to deal with a large asymmetric distribution of angle of incidence (AOI) as posed by high numerical aperture (NA) EUV projection optics. A six-mirror projection optics with an NA of 0.26 has been considered to design multilayer mirrors by rigorous investigation of pupil transmittance uniformity. For mirror surfaces having rotational asymmetry and a large inter-field span in AOI (the second but last mirror surface), the conventional uniform and radially graded Mo/Si-based periodic thickness mirrors (PTM) are found to fail to achieve appreciable pupil uniformity. 2D distributed PTM design, which satisfies Bragg criteria pointwise, also fails to achieve pupil uniformity due to large inter-field span and rotational asymmetry in AOI. To cope with such a surface, an approach with a laterally graded and vertically aperiodic thin film structure has been developed. This method is capable of achieving high pupil transmittance uniformity (standard deviation ∼1.95%) and average pupil transmittance (∼65.5%). The demonstrated method can also be applied to beyond extreme ultraviolet projection optics around 7 nm wavelength for next-generation semiconductor chip manufacturing technology.

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1 Introduction

In the present decade, extreme ultraviolet (EUV) lithography of NA 0.33 at 13.5 nm is primarily used for mass-scale semiconductor-device manufacturing.[1,2] Higher NA (0.55) lithography system has recently been demonstrated by ASML and is presently at the onset of mass-scale fabrications.[3,4] Due to its lower wavelength, EUV lithography has an edge over previously widely used ArF immersion lithography in achieving a better resolution limit. Due to the high absorbance of all the materials at such low wavelengths, the EUV projection optics rely on the multilayer reflective mirror. Unfortunately, conventional periodic multilayer mirrors (Mo/Si) exhibit high reflectance only in a limited angular bandwidth (∼12 deg near the normal angle of incidence[5]), which further decreases at higher angles of incidence. This poses a serious limitation on the achievable high NA of EUV projection optics as they rely on a higher angle of incidence. To reduce the angular load, high NA EUV projection optics involve central obscuration.[6,7] Central obscuration allows one to achieve high-NA within a relatively low AOI. However, having central obscuration is a compromise in the projection optics as it reduces the frequency

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response of the system at low frequencies and hence lowers the usable resolution for lithography.[8]

In the present work, considering a six-mirror projection optics of NA 0.26, we developed several design schemes for Mo/Si-based multilayer mirrors to cope with the complex angular distribution of AOI in such optics and thereby to achieve high pupil transmittance uniformity. The pupil-fill perfection is very crucial in determining the performance of lithography.[9–10] The field-averaged AOI and the inter-field deviation in AOI on all the reflective surfaces have been estimated by developing a CODE V (version 2025.03)[11] -based subroutine. Two kinds of design principles have mainly been explored, namely, (I) design by modifying thickness distribution on the horizontal plane (x–y) and (II) design by modifying both horizontal (x-y) and vertical (z) thickness distribution. The simulated results suggest that for mirror surfaces having rotational symmetry in angular distribution and having a small inter-field span in AOI, the first approach is sufficient to bring uniform pupil transmittance uniformity. On the other hand, for the projection optics involving a large rotationally asymmetric distribution and large inter-field deviation in the angle of incidence, the first approach fails, whereas the second approach in conjunction with horizontal averaging of AOI turns out to be very effective. This novel design approach using laterally graded and vertically aperiodic broadband mirrors can also be applied to the optics of beyond EUV (BEUV around 7 nm wavelength) projection system, which are currently under consideration for next-generation lithography.

2 Simulations and Results

An aberration-minimized six-mirror projection optics with image NA 0.26 and magnification ¼, previously patented by Nikon Corporation,[12] has been explored for present simulations as schematically displayed in Fig. 1(a). All the reflective surfaces (surface-1, 2, 4, 5, 6 & 7) are aspheric in nature, and the sag of the surface parallel to z-axis is given as follows:[12]

$$z = \frac{(y^2/R)}{1+\sqrt{1-y^2/R^2}} + C_4 y^4 + C_6 y^6 + C_8 y^8 + C_{10} y^{10} + C_{12} y^{12} + C_{14} y^{14} + C_{16} y^{16}, \quad (1)$$

where R is the apex radius of curvature and $C_4, C_6, C_8, C_{10}, C_{12}, C_{14}$, and C_{16} are the deformation coefficients of the surface. Surface-3 of the projection optics represents the aperture. Table 1 displays the values of all these aspheric surface parameters as well as the inter-surface separation (d). 10 representative field points on a fan-shaped image [Fig. 1(a)] have been considered for the purpose of calculating the distribution of the angle of incidence on all the reflective surfaces. The

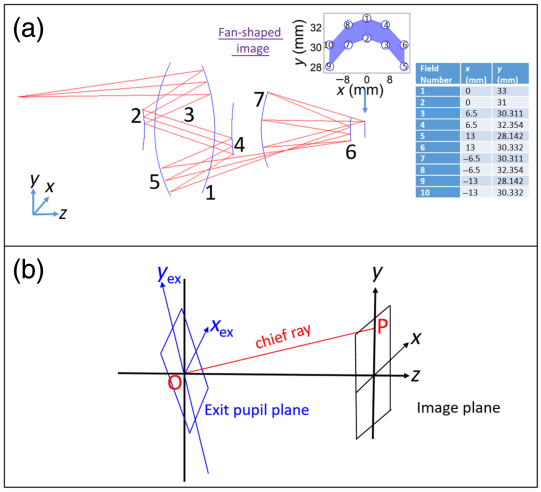


Fig. 1 (a) Six-mirror (surface-3 refers to aperture) EUV projection optics with a fan-shaped image (the inserted table displays the Cartesian coordinates of the 10 field points being considered for present simulations). (b) Schematic of pupil coordinate frame (x_{ex}, y_{ex}) as drawn on the perpendicular plane of the chief ray at the location of exit pupil (the axes on the pupil plane are the direction cosines of the axes on the image plane).

Table 1 Various parameters of six-mirror projection optics [Fig. 1(a)] are taken from Ref. 12, p. 15. (Inter-surface separation, d denotes the distance between the concerned surface and the next surface; +Ve and –Ve sign refer to $+z$ and $-z$ direction, respectively.)

Surface number	Radius of curvature (R) in mm	Inter-surface separation (d) in mm	C_4	C_6	C_8	C_{10}	C_{12}	C_{14}	C_{16}
			—	—	—	—	—	—	—
1	–615.08309	–261.993248	1.24222×10^{-10}	6.35317×10^{-16}	-1.06888×10^{-20}	2.64113×10^{-25}	-3.57081×10^{-30}	2.70722×10^{-35}	-8.78522×10^{-41}
2	–491.73881	140.083695	7.00678×10^{-09}	-5.15524×10^{-14}	2.48981×10^{-18}	-3.28561×10^{-22}	3.52305×10^{-26}	-2.47613×10^{-30}	8.34893×10^{-35}
3 (aperture)	∞	185.664836	—	—	—	—	—	—	—
4	714.67464	–287.402507	-8.87037×10^{-09}	1.19227×10^{-13}	-2.31508×10^{-18}	6.41111×10^{-23}	-3.11605×10^{-27}	1.15499×10^{-31}	-1.35301×10^{-36}
5	541.91375	729.071218	-6.30469×10^{-11}	-8.13359×10^{-16}	1.2336×10^{-20}	-3.03008×10^{-25}	3.81201×10^{-30}	-2.81098×10^{-35}	8.66299×10^{-41}
6	404.26328	–331.262443	2.06736×10^{-08}	3.31366×10^{-13}	1.34659×10^{-17}	-3.91416×10^{-21}	2.69954×10^{-24}	-9.53516×10^{-28}	1.39935×10^{-31}
7	411.78301	385.919779	6.28073×10^{-11}	7.36585×10^{-16}	5.04013×10^{-21}	-2.48786×10^{-26}	4.02094×10^{-30}	-1.46049×10^{-34}	2.35433×10^{-39}
Image	∞	—	—	—	—	—	—	—	—

Cartesian coordinates of the considered field points are tabulated in Fig. 1(a). Out of these, one representative field point, namely, field1$(x = 0$ mm$, y = 33$ mm$)$ has been considered for calculating the transmittance distribution on the exit pupil plane. The pupil plane and the corresponding coordinate frames are schematically displayed in Fig. 1(b). At first, the angle of incidence (AOI) for all the rays corresponding to a given field point and incident on a given reflective surface has been calculated using CODE V[11] analysis software. Subsequently, at every location on the reflective surface, the average of AOI (ϕ) and the inter-field span of AOI ($\Delta\phi$) including all 10 field points have been computed employing a subroutine written within macro-plus of CODE V. The simulated average of AOI (ϕ) and the inter-field span of AOI ($\Delta\phi$) are plotted in polar coordinate in Figs. 2 and 3, respectively, using a color map.

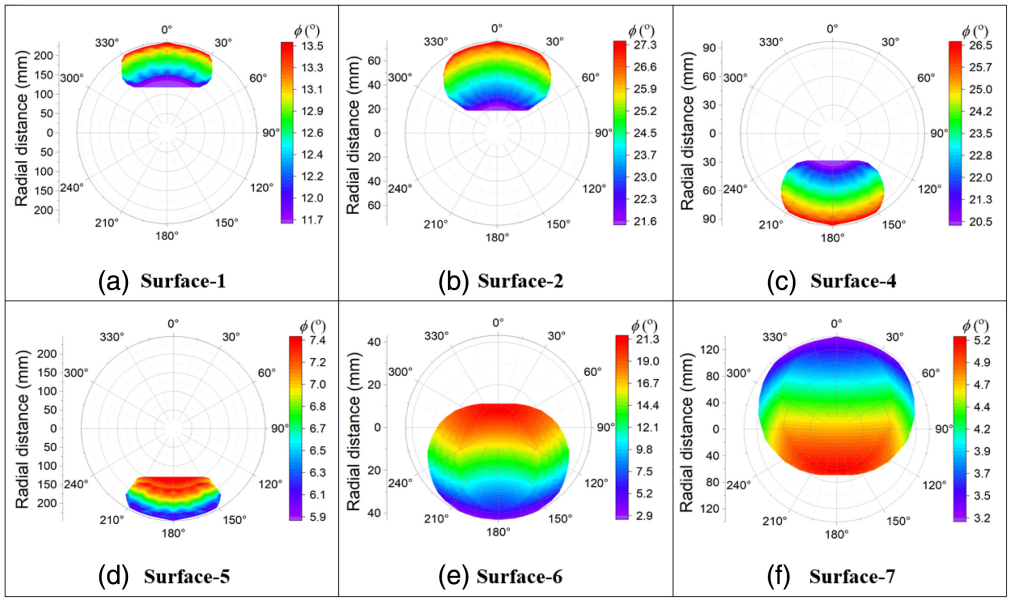


Fig. 2 Color maps displaying the distribution of field averaged AOI, ϕ on various surfaces of the 6-mirror projection optics.

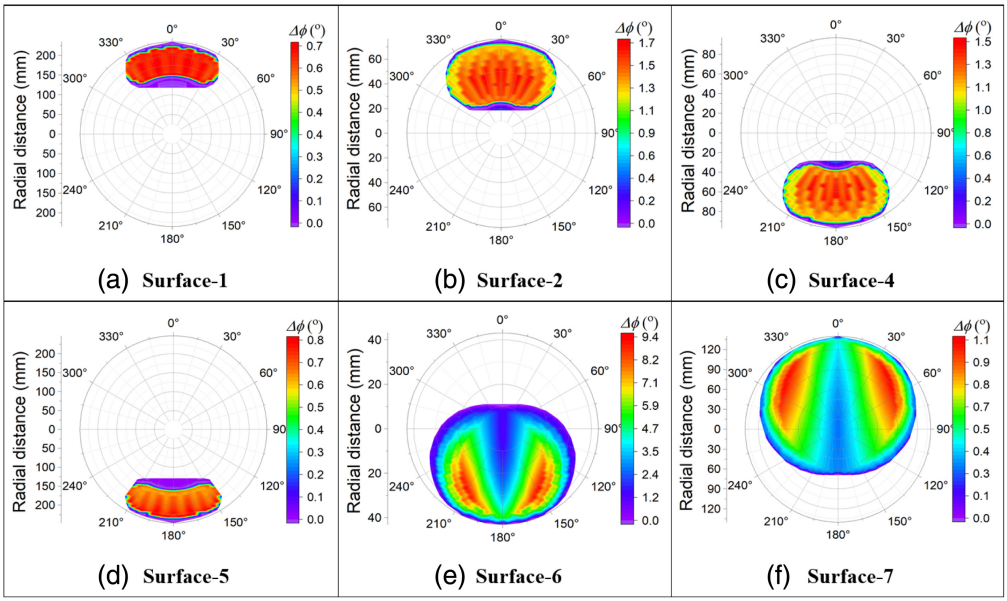


Fig. 3 Color maps displaying inter-field angular span, $\Delta\phi$ on various surfaces of the six-mirror projection optics.

From Figs. 2 and 3, it can be observed that

1. Unlike other surfaces, surface-6 and surface-7 exhibit the distribution of ϕ, which is rotationally asymmetric.
2. $\Delta\phi$ is significantly low (<1.73 deg) for all the surfaces except that for surface-6. The typical angular acceptance for Mo/Si multilayer mirror[5] is greater than this angular span. Thus, the variation of angle of incidence among the fields does not pose a greater challenge in designing multilayer mirror structures for the surfaces other than surface-6.

For both these reasons, the designing of a multilayer mirror structure for surface-6 becomes extremely challenging. In Sec. 2.1, we present three designs schemes for surface-6, namely, (i) uniform PTM (ii) radially graded PTM, and (iii) 2D distributed PTM by modifying the horizontal thickness distribution of the Mo/Si multilayer structure.[13] In Sec. 2.2, we have explored a novel approach by adopting a horizontal averaging of AOI and modifying both horizontal and vertical thickness distribution of multilayers to cope with the aforementioned large $\Delta\phi$ and asymmetric distribution in AOI for surface-6. In all cases, the multilayer mirror has been applied on surface-6, assuming the rest of the surfaces to be 100% reflective to simulate the transmittance maps on the exit pupil plane [Fig. 1(b)].

2.1 Design by Modifying Horizontal (x–y) Thickness Distribution

2.1.1 *Uniform PTM*

Under this scheme, a periodic thickness Mo/Si multilayer mirror structure (alike[14]) has been designed comprising 41 bilayers. The reflectivity spectrum has been simulated using the "IMD" code[15] under the "XOP" (X-ray Oriented Program) software package following the Parrat's formalism.[16] To decide the periodic thickness (sum of Mo and Si layer thicknesses) of the bilayers, field-averaged AOI, ϕ [Fig. 2(e)] of the surface-6 has further been radially averaged to obtain the radial angle of incidence, $\Phi(r)$ as plotted in Fig. 4(a) as a function of radius, r. As indicated in Fig. 4(a), Φ_{r_1} and Φ_{r_2} represent the AOI at two extreme radii r_1 and r_2, while Φ_c represents the central AOI ($\Phi_c = (\Phi_{r_1} + \Phi_{r_2})/2$) of the surface. Thus, the mirror to be developed should exhibit peak reflectivity at angle Φ_c (10.84 deg). The value of Φ_c has been used to optimize the periodic thickness of the Mo and Si layer in multilayer structures. The thickness ratio of Mo and Si layer as defined by Γ (Mo thickness/(Mo thickness + Si thickness) has been kept at 0.4 to maximize reflectivity.[17] The obtained optimized periodic thickness of the mirror for surface-6 is 6.99 nm, which is uniform across the radius, as plotted in Fig. 4(a). The obtained reflectivity spectrum at 13.4 nm wavelength has been plotted in Fig. 4(b) along with the schematic of the optimized multilayer structure in the inset. Applying such a mirror on surface-6, we simulated the pupil transmittance map, which is displayed on Fig. 4(c).

2.1.2 *Radially graded PTM*

Under this scheme, a laterally graded periodic thickness Mo/Si mirror structure (alike[18]) of 41 bilayers is designed so that the peak angle of reflectivity spectrum shifts from center to periphery as per the requirement. The thickness of the comprising layer at any location on the surface is rotationally symmetric with respect to center of the surface. In this scheme, the periodic thickness, D_c corresponding to peak at central AOI, Φ_c is initially estimated following the method as described in design (i). Subsequently, the periodic thickness, $D(r)$ at any given radius, r is estimated as per following method using Bragg condition:[19]

$$2D_c \cos \Phi_c = \lambda \tag{2}$$

and

$$2D(r) \cos \Phi(r) = \lambda \tag{3}$$

Combining Eqs. (2) and (3), we get

$$D(r) = D_c \times \frac{\cos \Phi_c}{\cos \Phi(r)} \tag{4}$$

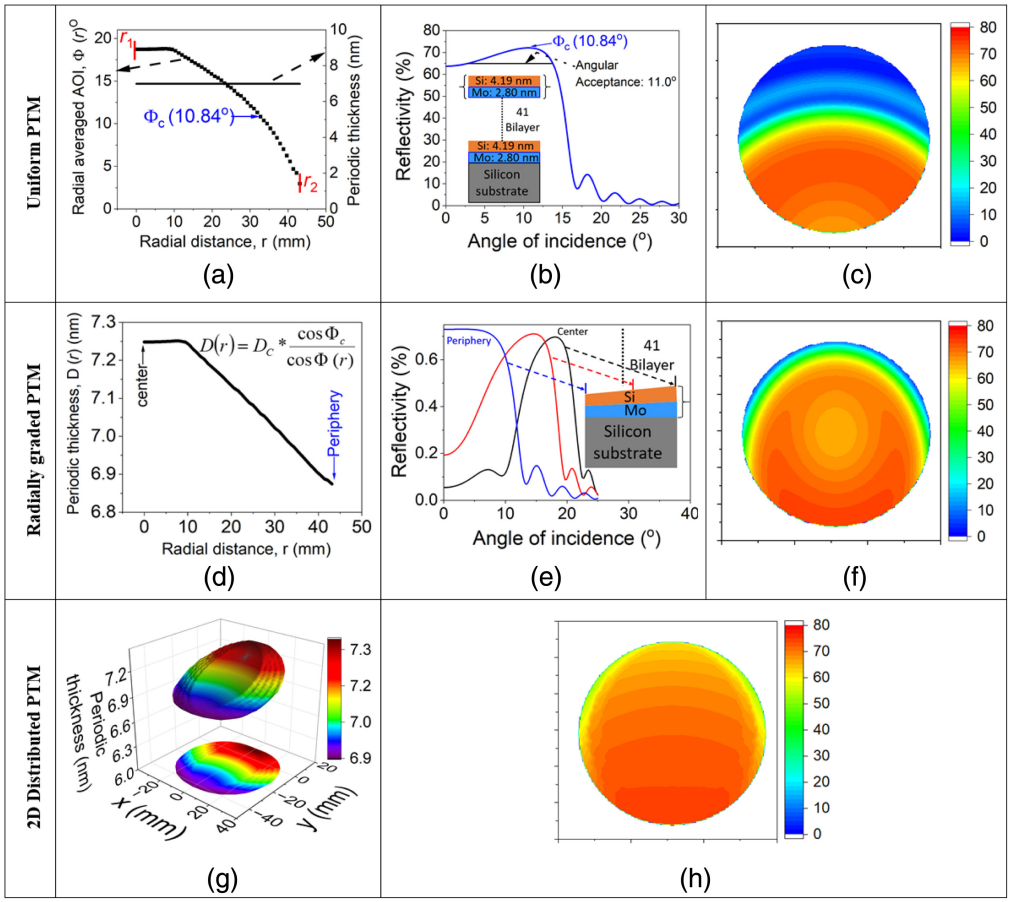


Fig. 4 (a) Radial averaged angle of incidence, $\Phi(r)$ and optimized periodic thickness for uniform PTM plotted as a function of radius, r; (b) reflectivity spectrum and the optimized mo/Si multilayer structure of uniform PTM for surface-6; (c) simulated pupil transmittance map applying uniform PTM on surface-6; (d) variation of periodic thickness of radially graded PTM for surface-6; (e) reflectivity spectra at three representative radial locations and the optimized Mo/Si multilayer structure of designed radially graded PTM; (f) simulated pupil transmittance map applying radially graded PTM on surface-6; (g) distribution of periodic thickness of 2D distributed PTM for surface-6; (h) simulated pupil transmittance map applying 2D distributed PTM on surface-6.

The estimated radius-dependent periodic thickness, $D(r)$ of the designed graded Mo/Si mirror structure for surface-6 is plotted in Fig. 4(d). The simulated reflectivity spectra at three representative radial locations are displayed in Fig. 4(e) along with the schematic of the graded multilayer structure in the inset. Applying this radially graded periodic thickness mirror on surface-6, the pupil transmittance map is simulated and plotted in Fig. 4(f).

2.1.3 *2D distributed PTM*

Under this scheme, a periodic Mo/Si multilayer (41 bilayers) mirror structure is designed in which periodic layer thickness at every location of the mirror is calculated pointwise based on the average of AOI (ϕ) for surface-6, shown in Fig. 2(e). At any given location (x, y), the periodic thickness of the mirror is directly estimated using the following Bragg condition [following Eq. (4)].

$$D(x, y) = D_c \times \frac{\cos \Phi_c}{\cos \phi(x, y)} \tag{5}$$

Essentially, in this scheme, the periodic thickness becomes a 2-dimensional function in the plane of the mirror. The estimated periodic thickness distribution on the horizontal plane (x–y) for surface-6 is plotted in Fig. 4(g), and the simulated pupil transmittance map applying this mirror is displayed in Fig. 4(h).

As can be observed from Figs. 4(c) and 4(f), the simulated pupil transmittance maps are not very uniform, and the standard deviation of pupil transmittance is ~28.61% and ~14.63% respectively. In the design scheme (i), the achievable angular acceptance [Fig. 4(b)] for the high reflecting regime of the mirror is less than the required span of AOI, ϕ across surface-6 [Fig. 2(e)], which makes the pupil transmittance map in Fig. 4(c) nonuniform. Design scheme (ii), which applies laterally graded mirror to cope with the lateral variation of AOI, also fails to bring pupil uniformity [Fig. 4(f)], due to the fact that surface-6 exhibits appreciable rotational asymmetry in distribution of AOI [Fig. 2(e)] and the required $\Delta\phi$ [Fig. 3(e)] is relatively large compared with the achievable angular acceptance. On the contrary, as presented in Appendix A, for the rest of the surfaces (surface-1, 2, 4, 5, and 7), design scheme (i) and (ii) suffice to achieve pupil uniformity predominately due to smaller $\Delta\phi$ (Fig. 3). In design scheme (iii) for surface-6, even though the periodic thickness of the multilayer structure has been accurately applied at every location without any radial approximation, the pupil transmittance map [Fig. 4(h)] still suffers from slight nonuniformity, and the estimated standard deviation is ~4.44%. This is due to the fact that inter-field angular span, $\Delta\phi$ for this surface is pretty large (0 deg to 9.4 deg) as displayed in Fig. 3(e), i.e., the AOI for individual field points largely deviates from the average AOI at any given location. This leads to the noncompliance of the Bragg condition for all individual field points and hence results in nonuniformity of the pupil transmittance map. Moreover, design (iii) involves two-dimensional distribution of comprising layer thicknesses of the mirror, and hence the fabrication procedure is very complex and subject to feasibility. In Sec. 2.2, we present a novel approach by modifying vertical thickness distribution of multilayers along with introducing lateral thickness gradient to cope with the aforementioned large $\Delta\phi$ and asymmetric distribution in AOI for surface-6.

2.2 Design by Modifying Both Horizontal (x–y) and Vertical (z) Thickness Distribution

Figure 5(a) displays the map of average AOI, ϕ for surface-6 as copied from Fig. 2(e). As it can be observed, the radial averaging of ϕ (along the blue circular line) at location P gives an underestimated average value of AOI due to rotational asymmetry. On the other hand, at location Q, radial averaging yields an overestimated average AOI in compared with the actual value. To get rid of such an error, we apply the scheme of horizontal averaging of AOI (along the horizontal purple line). Under this scheme, the AOI, at a given location "P," is approximated as the average AOI along the imaginary horizontal line AB passing through location "P" as displayed in Fig. 5(a). A MATLAB-based code has been written to estimate horizontally averaged AOI, $\Phi(y)$ through interpolation. The estimated horizontally averaged AOI, $\Phi(y)$ and radially averaged AOI, $\Phi(r)$ are simultaneously plotted in Fig. 5(b) for easy comparison. It can be observed that

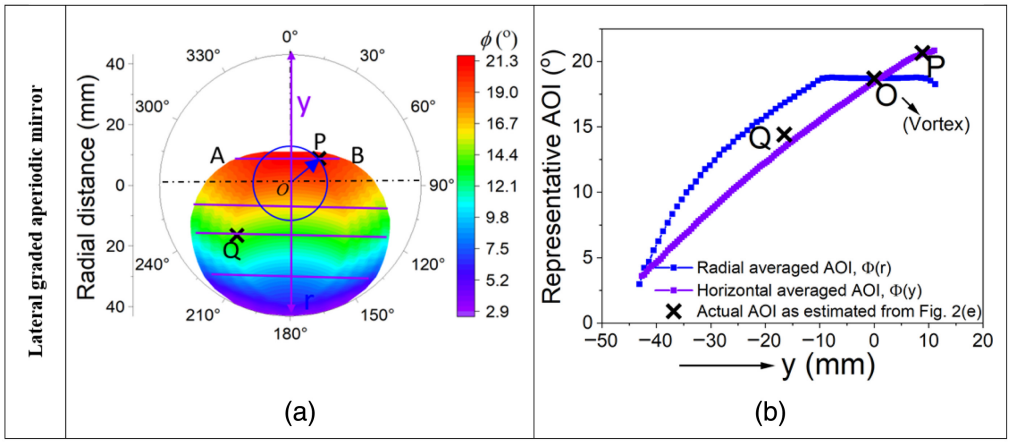


Fig. 5 (a) Schematic for calculation of radially averaged (along blue circular line) AOI, $\Phi(r)$ and horizontally averaged (along purple horizontal line) AOI, $\Phi(y)$ using field averaged AOI, ϕ in Fig. 2(e) for surface-6; (b) plot of estimated $\Phi(y)$ and $\Phi(r)$ as a function of ordinate, y, for surface-6. "X" mark indicates the actual AOI as estimated from Fig. 2(e) for a few representative locations.

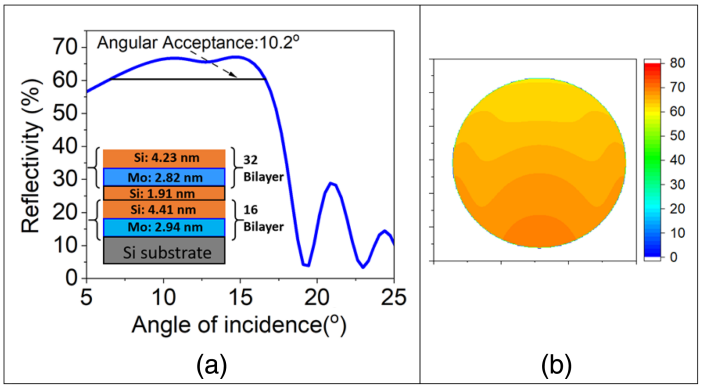


Fig. 6 (a) Reflectivity spectra for the designed aperiodic broadband mirror for surface-6 (the optimized multilayer structure is inserted in the inset); (b) pupil transmittance map for surface-6 assuming lateral graded aperiodic mirror.

at a representative location, P, the horizontal averaged AOI ($\Phi(y) = 20.63$ deg) is closer to the actual AOI (20.65 deg) than that of the radial averaged AOI ($\Phi(r) = 18.75$ deg) proving the appropriateness of the former approximation in compared with later.

We designed an aperiodic Mo/Si multilayer structure-based mirror exhibiting a large angular bandwidth of the reflectance peak. The optimized design and the reflectivity curve are plotted in Fig. 6(a). The angular acceptance in which the reflectivity remains greater than 90% of peak reflectivity is ~10.2 deg. This base design has been applied as a reflector on surface-6 by introducing lateral gradiance in layer thickness based on horizontally averaged AOI, $\Phi(y)$. At first, the central AOI, Φ_c has been estimated from the range of $\Phi(y)$ following the method as described in design (i). Then the optimized thickness, d_c of a layer has been estimated so that the reflection peak of the aperiodic mirror occurs at the central AOI, Φ_c. Subsequently, the thickness, $d(y)$ of the layer at ordinate y has been calculated using the condition [following Eq. (4)]:

$$d(y) = d_c \times \frac{\cos \Phi_c}{\cos \Phi(y)} \tag{6}$$

This lateral *graded* aperiodic mirror has been applied on surface-6 and the obtained pupil transmittance map is displayed in Fig. 6(b). This can be observed that the pupil transmittance map achieved by applying such a mirror is fairly uniform (standard deviation ~1.95%). Moreover, in this design scheme, the layer thicknesses are one-dimensionally graded and hence can be easily fabricated. Such a laterally graded aperiodic mirror can be fabricated by the shadow-mask magnetron sputtering technique[20–22] or with a rotation and revolution mechanism[23] after some minor mechanical modifications.

3 Summary & Conclusion

In this paper, Mo/Si-based multilayer EUV mirror has been designed for an aberration minimized six-mirror projection optics set-up.[12] The distributions of the angle of incidence on all the reflecting surfaces have been computed through sequential calculations. Several schemes of Mo/Si multilayer structure have been considered to design mirrors for these reflecting surfaces to achieve both high pupil transmittance uniformity and high pupil transmittance magnitude. Four numbers of design schemes have been explored, namely, (i) uniform PTM, (ii) radially graded PTM, (iii) 2D distributed PTM, and (iv) lateral graded aperiodic mirror. The findings suggest, for reflective surface-1, 5, and 7 the design scheme (i) is recommended (Appendix A) as it can achieve high pupil uniformity, and the scheme is simple from both design and fabrication points of view. On the other hand, for reflective surfaces 2 and 4, (ii) radially graded thickness Mo/Si multilayer mirrors suffice to achieve pupil uniformity and hence are recommended (Appendix A). However, for surface-6, which suffers from large inter-field angular deviation and rotationally asymmetric distribution of AOI, the design of mirror structure becomes very challenging. Table 2 summarizes the result of average pupil transmittance, $\langle T \rangle$ and standard deviation of pupil transmittance, ΔT for surface-6 as obtained by applying all four designs as

Table 2 The pupil transmittance map for surface-6 has been summarized under various design schemes of multilayer mirror structures.

	Average transmittance $\langle T \rangle$ (%)	Standard deviation (ΔT) (%)
(i) Uniform PTM	45.14	28.61
(ii) Radially graded PTM	63.43	14.63
(iii) 2D distributed PTM	69.37	4.44
(iv) Lateral graded aperiodic mirror	65.5	1.95

mentioned above. In Appendix B, the average pupil transmittance, $\langle T \rangle$ and standard deviation of pupil transmittance, ΔT have been compared for two representative field points, namely, field-1 and field-10. As these field points are located in the same radius and are just azimuthally offset, the multilayer mirror structures of design (i) and (ii), which are radially symmetric, yield similar pupil performance. On the other hand, the multilayer mirror structures of design (iii) and design (iv) are not radially symmetric. Hence, their pupil performance slightly deteriorates as the field point is shifted away from y axis ($x = 0$). As can be observed from Table 2, both designs (iii) and (iv) may be considered to be effective, yielding low standard deviation. Design (iii) gives relatively higher pupil transmittance but lower pupil uniformity (i.e., larger ΔT) compared with design (iv). So there exists a slight tradeoff between the pupil transmittance and pupil uniformity between these two design schemes. Second, design (iii) involves two-dimensional distribution of comprising layer thicknesses of the mirror, and hence the fabrication procedure is very complex and subject to feasibility. By contrast, in design (iv) the layer thicknesses are one-dimensionally graded and hence can be easily fabricated. Such a laterally graded aperiodic mirror can be fabricated by the shadow-masks magnetron sputtering technique[20–22] or with a rotation and revolution mechanism[23] after some minor mechanical modifications. Overall, the results successfully demonstrate an approach to achieve high pupil transmittance uniformity (standard deviation ~1.95%) even for the surface having rotational asymmetry and a large inter-field span of angle of incidence. The demonstrated method can be extended to higher NA systems at 13.5 nm wavelength. Depending upon the optical configuration of the high NA projection system, the rotational asymmetry in average AOI and large inter-field span of angle of incidence may occur for one or more than one surface, by contrast to only one surface (the second but last surface) in this work. Accordingly, design (iii) or (iv) may be appropriate for such a surface to achieve pupil uniformity. The demonstrated method can also be applied to beyond extreme ultraviolet (BEUV) projection optics around 7 nm wavelength[24] for next-generation semiconductor chip manufacturing technology. By contrast to 2-periods lateral graded aperiodic mirror in the EUV region, in the BEUV region, a complicated multilayer structure with a larger number of layers may be exploited to enhance the angular bandwidth of the mirror and to achieve high pupil transmittance uniformity. Moreover, 2D thickness distribution in conjunction with an aperiodic broadband mirror may be an effective approach to achieve high pupil transmittance uniformity in the BEUV wavelength region. Such attempts to achieve high pupil transmittance uniformity in the BEUV wavelength region will be made in next step. Multilayer models incorporating interdiffusion layer (for example, $MoSi_2$) can be incorporated in these attempts in order to represent more realistic condition.

4 Appendix A

For the six-mirror projection optics, displayed in Fig. 1(a), the pupil transmittance maps have been simulated by applying Mo/Si multilayer structure-based (i) uniform PTM and (ii) radially graded PTM on each surface individually. As mentioned in the paper, all the pupil maps have been simulated for a representative field points (field number 1) at 13.4 nm wavelength of EUV light. The rigorous method for designing mirrors under the above two design schemes has been

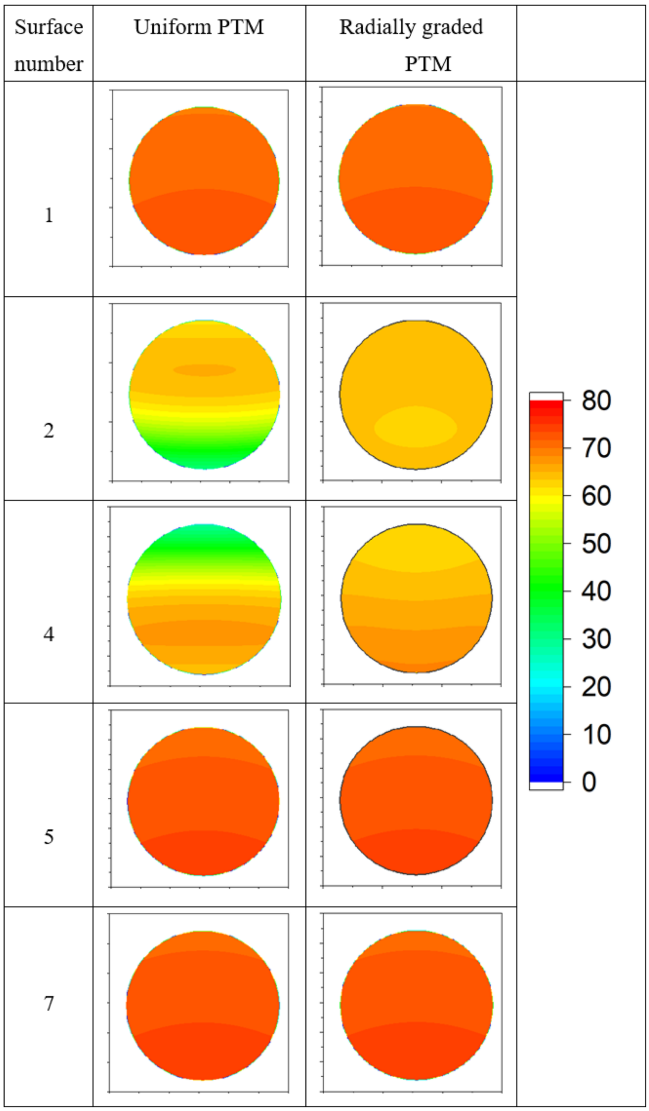


Fig. 7 Pupil transmittance map simulated applying (i) uniform PTM and (ii) radially graded PTM on the individual surface, assuming rest of the surfaces to be 100% reflective.

presented in the paper for a representative surface (surface-6). The similar methods have been applied for all the remaining surfaces (surface-1, 2, 4, 5, and 7), and the obtained pupil transmittance maps are displayed in this document in Fig. 7. Surface-3 represents the aperture of the projection optics for which pupil maps have not been simulated. To quantify the performance of the pupil maps, the average pupil transmittance, $\langle T \rangle$ and standard deviation of pupil transmittance, ΔT have been estimated and tabulated in Table 3.

As can be observed from Table 3, the pupil transmittance maps obtained using uniform PTM are pretty uniform for surface-1, 5, and 7. To understand the relationship between the field-averaged angle of incidence (AOI), ϕ and the pupil uniformity of the respective surfaces, the lateral variation of ϕ across each surface (as obtained from Fig. 2) is mentioned in Table 3. As can be observed from Table 3, the lateral variation ϕ for surface-1, 5, and 7 is relatively less. For this reason, the achievable angular acceptance using uniform PTM becomes sufficient to cover this small variation in angle of incidence over the surface. Hence, a fairly uniform pupil map ($\langle T \rangle = 0.78\%$ to 0.86%) could be achieved for these surfaces with uniform PTM. On the other hand, for surface 2 and 4, there exists large lateral variation of field averaged AOI, ϕ across the surface. Thus, the achievable angular acceptance with uniform PTM lacks to cover this large angular variation and hence leads to larger standard deviation in pupil transmittance ($\langle T \rangle = 8.62\%$ to 10.32%). However, when radially graded PTMs are applied on these two

Table 3 Average pupil transmittance, $\langle T \rangle$ and standard deviation of pupil transmittance, ΔT for various surfaces of projection optics under two design schemes. (Column 4 tabulates the lateral variation of field averaged AOI, ϕ for all surfaces as estimated from Fig. 2.)

Surface number	Uniform PTM	Radially graded PTM	Lateral variation of ϕ across surface
1	$\langle T \rangle = 71.39\%$; $\Delta T = 0.78\%$	$\langle T \rangle = 71.42\%$; $\Delta T = 0.79\%$	11.73 deg to 13.53 deg
2	$\langle T \rangle = 58.38\%$; $\Delta T = 8.62\%$	$\langle T \rangle = 63.89\%$; $\Delta T = 0.43\%$	21.62 deg to 27.28 deg
4	$\langle T \rangle = 58.55\%$; $\Delta T = 10.32\%$	$\langle T \rangle = 65.12\%$; $\Delta T = 1.85\%$	20.52 deg to 26.48 deg
5	$\langle T \rangle = 72.6\%$; $\Delta T = 0.83\%$	$\langle T \rangle = 72.62\%$; $\Delta T = 0.82\%$	5.92 deg to 7.40 deg
7	$\langle T \rangle = 72.86\%$; $\Delta T = 0.86\%$	$\langle T \rangle = 72.86\%$; $\Delta T = 0.85\%$	3.23 deg to 5.15 deg

surfaces, due to the angle tunability of the reflectance peak across the lateral direction, standard deviation of the pupil transmittance map (0.43% to 1.85%) could be greatly reduced.

In summary, design (i) based on Mo/Si multilayer structure can achieve high pupil uniformity for reflective surfaces-1, 5, and 7. Hence, for these three surfaces, design (i) may be recommended due to simplicity in both design and fabrication points of view. On the other hand, for reflective surfaces 2 and 4, design (ii) is recommended to achieve uniform pupil transmittance.

5 Appendix B

So far all the pupil transmittance maps have been presented for field-1($x = 0$ mm, $y = 33$ mm) only. Figures 4(c), 4(f), 4(h), and 6(b) display the obtained pupil transmittance maps corresponding to Field-1 for four different designs explored in the paper. To investigate the effect of variation of field points, the pupil transmittance maps have further been simulated for field-10 ($x = -13$ mm, $y = 30.332$ mm) for the said designs of multilayer mirror structures. Figure 8

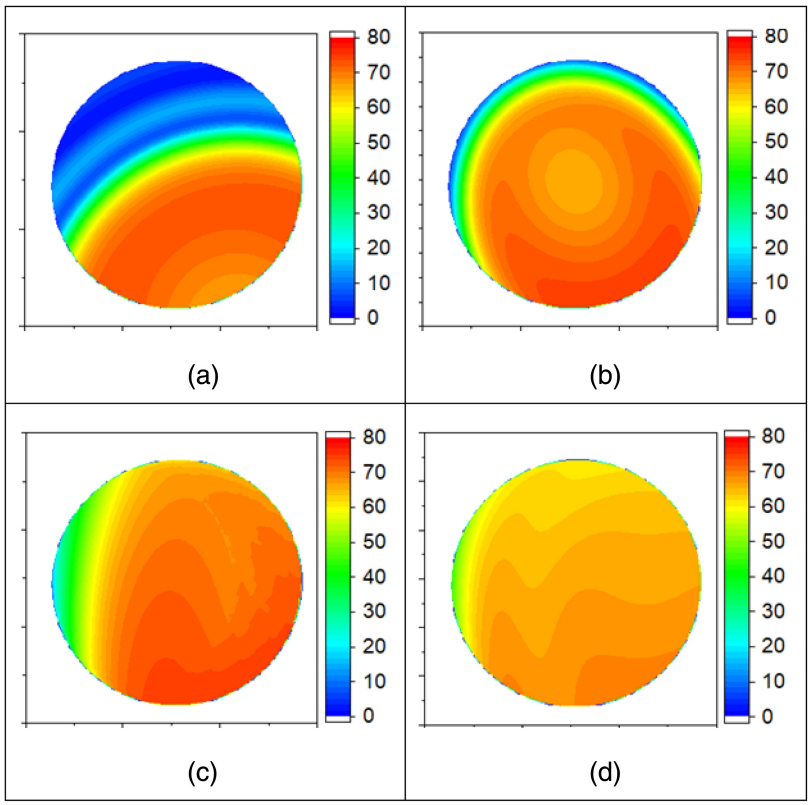


Fig. 8 Simulated pupil transmittance map for field-10 applying (a) uniform PTM, (b) radially graded PTM, (c) 2D distributed PTM, and (d) lateral graded aperiodic mirror on surface-6.

Table 4 Summary of performance of pupil transmittance map for surface-6 for field-1 and field-10 under various design schemes of multilayer mirror structures.

	Field-1		Field-10	
	Average transmittance $\langle T \rangle$ (%)	Standard deviation (ΔT) (%)	Average transmittance $\langle T \rangle$ (%)	Standard deviation (ΔT) (%)
(i) Uniform PTM	45.14	28.61	45.13	28.62
(ii) Radially graded PTM	63.43	14.63	63.41	14.65
(iii) 2D distributed PTM	69.37	4.44	66.71	9.31
(iv) Lateral graded aperiodic mirror	65.5	1.95	65.14	3.08

displays the obtained pupil transmittance maps. To quantify the performance of the pupil maps, the average pupil transmittance, $\langle T \rangle$ and standard deviation of pupil transmittance, ΔT have been estimated and tabulated in Table 4. For the sake of comparison, $\langle T \rangle$ and ΔT for field-1 (copied from Table 1) are also appended in Table 4. As can be observed from Table 4, applying design (i) and design (ii) of multilayer mirror yields almost similar values of $\langle T \rangle$ and ΔT for both the field points 1 & 10. However, applying design (iii) and design (iv) of multilayer mirror yields slightly inferior quality (lower $\langle T \rangle$ and higher ΔT) of pupil performance for field-10 as compared to field-1. As the field-1 and field-10 are located in the same radius and are just azimuthally offset, the multilayer mirror structures of design (i) and (ii), which are radially symmetric, yield similar pupil performance. On the other hand, the multilayer mirror structures of design (iii) and design (iv) are not radially symmetric. Hence, their pupil performance slightly deteriorates as the field point is shifted away from y axis ($x = 0$).

Disclosures

The authors declare no conflicts of interest.

Code and Data Availability

There is no supporting data for this paper. The associated code has been uploaded to Code Ocean.

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